Beyond Reward Suppression: Reshaping Steganographic Communication Protocols in MARL via Dynamic Representational Circuit Breaking

Liu Hung Ming*
cyril.liu@gmail.com

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Abstract

*PARRAWA AI

1 Methods

1.1 Dataset and Experimental Design

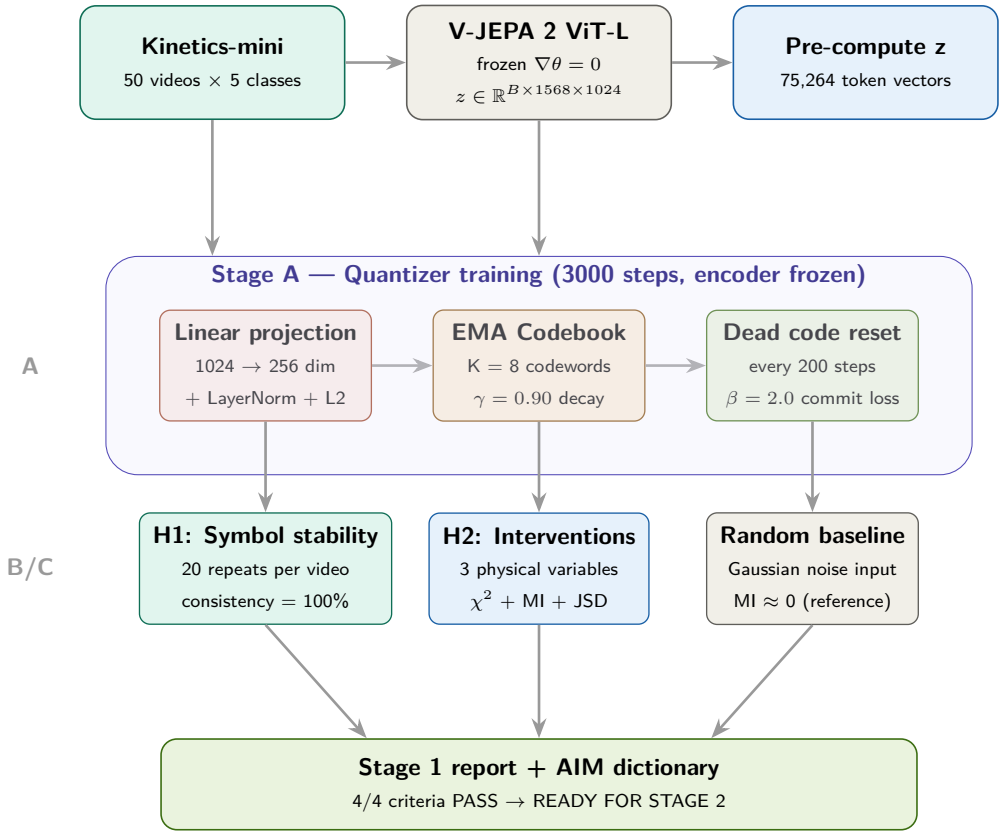

Figure 1: Stage 1 pipeline overview. *Top row* (A): The input pipeline feeds 50 Kinetics-mini videos through the frozen V-JEPA 2 ViT-L encoder ($\nabla\theta = 0$), producing $B \times 1568 \times 1024$ latent token vectors that are precomputed once and cached. *Middle row* (A): The Stage A quantizer trains for 3,000 steps on the precomputed vectors using a linear projection layer (1024 → 256 dimensions, followed by LayerNorm and L2 normalization), an EMA-updated codebook ($K = 8$, $\gamma = 0.90$), and a dead-code reset mechanism ($\beta = 2.0$ commitment loss); the V-JEPA 2 encoder weights remain fully frozen throughout. *Bottom row* (B/C): After training, the pipeline is evaluated on two diagnostic criteria: H1 symbol stability (20 repeated forward passes per video; consistency = 100%) and H2 intervention experiments (three physical-variable pairs assessed via χ^2, MI, and JSD), with a Gaussian-noise random baseline confirming MI ≈ 0 for unstructured inputs. All four pass criteria are satisfied, producing the Stage 1 report and AIM dictionary that certify readiness for Stage 2.

Figure 1 provides an overview of the complete Stage 1 pipeline, from data ingestion through quantizer training to diagnostic evaluation.

Dataset. We use the validation split of the Kinetics-mini dataset, selecting five action categories: *archery*, *bowling*, *flying_kite*, *high_jump*, and *marching*, with at most 10 videos per category. Of these, 48 videos ultimately entered training (see Section 1.2 for details). Each video is sampled at $T = 16$ frames at 224×224 resolution, processed through V-JEPA 2's official processor with center cropping and ImageNet normalization before being passed to the encoder.

Rationale for the Intervention Design. The central question of Stage 1 is whether AIM's symbolization mechanism can read structured information from V-JEPA 2's frozen latent space. To investigate this, we design controlled intervention experiments: all other conditions are held fixed while a single physical dimension is varied, and we observe whether the AIM symbol distribution changes in a statistically significant manner.

Kinetics-mini is an action-classification dataset whose labels denote action categories rather than physical attributes. It is therefore impossible to directly manipulate a single physical variable (e.g., holding scene, subject, and action constant while varying only grasp angle). Instead, we adopt a **category-proxy** strategy: we select pairs of action categories that exhibit large natural differences along one target physical dimension while minimizing differences along other major visual dimensions, using inter-category contrast as a proxy for physical-variable contrast.

The validity of this strategy rests on the following premise. V-JEPA 2 is pretrained to predict masked spatiotemporal regions in latent space, so its encoder must compress video into structured representations that support spatiotemporal prediction. Such representations inevitably encode physical attributes such as object shape, pose, and motion pattern. When two action categories differ substantially along one physical dimension, that difference should leave a measurable directional shift in latent space. Stage 1 tests whether this shift is sufficiently pronounced to be captured by AIM's discrete symbolization mechanism. Kinetics-mini serves as a publicly reproducible benchmark providing a verifiable starting point.

Two practical reasons motivate the use of an existing video dataset rather than purpose-built controlled footage. First, the goal of this stage is *diagnostic*: to verify architectural compatibility between AIM and V-JEPA 2, not to precisely quantify the encoding strength of specific physical variables. Category proxies are sufficient for the statistical-significance tests that serve this purpose. Second, acquiring video data with fine-grained control over individual physical variables requires dedicated data-collection infrastructure that exceeds the resource scope of Stage 1.

Category Pair Selection. The five categories are grouped into three contrastive pairs, each targeting one physical dimension. Pairs were chosen to maximize inter-category differences along the target dimension while minimizing differences along other salient visual features (e.g., scene type, number of persons, range of motion), thereby reducing confounding variables.

- **Grasp angle.** *Archery* (drawing a bow: three-finger pinch grip on the string, arm extended into a static release posture with shoulder rotation) vs. *bowling* (single-hand grip through finger holes, dynamic forward-swing throw). Both involve a single person, a hand-held object, and upper-limb-dominated motion; the primary differences are grip morphology and the direction of arm force application. We note that scene environment (outdoor archery range vs. indoor bowling lane) and background color also differ—an unavoidable confound of the category-proxy strategy.

- **Object geometry.** *Flying_kite* (an elongated linear object controlled indirectly via string) vs. *high_jump* (no manipulated object; the subject's body executes an arching leap). The two

conditions differ markedly in the geometry of the primary object and in the spatial relationship between the human body and that object.

- **Motion speed / temporal structure.** *Marching* (periodic, regular gait with a $\approx$2 Hz repetitive temporal structure) vs. *archery* (near-static loading followed by a single rapid release; aperiodic temporal structure). Because V-JEPA 2's pretraining objective directly involves temporal prediction, we expect it to be most sensitive to temporal-structure differences—a prediction that is consistent with the experimental results, in which the motion-speed intervention yields the highest MI and JSD of the three pairs (Section 2.3).

Reuse of Archery. *Archery* appears in both the grasp-angle pair (contrasted with bowling) and the motion-speed pair (contrasted with marching). This is a design compromise imposed by the limited dataset scale, but it simultaneously provides an additional validation opportunity: if AIM's symbol system captures stable latent features of the action category rather than incidental statistical artifacts introduced by the intervention framework, then *archery* should produce consistent symbol distributions across the two independent interventions. The experimental results confirm this (Section 2.3).

Limitations of the Category-Proxy Strategy. The fundamental limitation of this approach is that the symbol-distribution differences we observe reflect the overall differences between the selected action categories in V-JEPA 2's latent space, not the isolated effect of the target physical variable. For the grasp-angle intervention, for instance, the latent-space distance between *archery* and *bowling* subsumes contributions from grasp angle, scene environment, object material, and other factors that the experimental design cannot disentangle. The correct characterization of our results is therefore: *AIM symbols have statistically significant discriminative power for action-category pairs whose primary difference lies along a specified physical dimension*—not that AIM symbols directly encode the target physical variable. Ideal follow-up validation should employ carefully controlled synthetic or robot-manipulation videos that isolate the target variable while holding all other visual factors constant; this is deferred to Stage 3 or future work.

1.2 Frozen V-JEPA 2 Encoder and Latent Space Precomputation

Scientific Necessity of the Freezing Strategy. The core scientific question of this stage is whether AIM's symbolization mechanism can read knowledge already acquired by V-JEPA 2. If the encoder were left unfrozen, two confounds would be introduced: (1) the encoder could actively adapt its output distribution in response to the quantizer, and (2) the quantizer could acquire separability through encoder gradients that would not otherwise exist. Freezing the encoder ensures that, for any input $\mathbf{x}$, the mapping $E_\phi(\mathbf{x})$ is constant and all parameter gradients are fully blocked:

$$\forall\, \theta \in \Phi : \quad \frac{\partial \mathcal{L}}{\partial \theta} = 0, \qquad E_\phi(\cdot)\big|_{\text{eval}}. \tag{1}$$

Freezing also yields a practical benefit: because a frozen encoder produces identical outputs for identical inputs, all latent vectors $\mathbf{z}$ can be precomputed once and reused throughout training, eliminating the need to re-execute an encoder forward pass at every training step.

Preliminary Latent Space Diagnosis. Prior to designing the quantizer, we executed to characterize the numerical properties of V-JEPA 2's output distribution across 5 representative videos. This diagnostic revealed the key challenge: the per-token L2 norm of 97.70 ± 0.81 is exceptionally large and highly uniform (coefficient of variation $< 0.6\%$), meaning that all latent vectors are nearly equidistant from any randomly initialized codebook entry. Direct quantization without normalization would therefore cause immediate codebook collapse, a finding that directly motivated the projection-plus-normalization pipeline described in Section 1.3.

Token Structure of V-JEPA 2 ViT-L. V-JEPA 2 uses tubelet embedding with temporal patch size $t_p = 2$ and spatial patch size $p = 16$. For $T = 16$ input frames the number of output tokens is

$$N_{\text{tokens}} = \frac{T}{t_p} \times \left(\frac{H}{p}\right)^2 = \frac{16}{2} \times \left(\frac{224}{16}\right)^2 = 8 \times 196 = 1{,}568. \tag{2}$$

Each tubelet token spans 2 frames in time and a 16×16 pixel patch in space. With embedding dimension $D = 1{,}024$, the encoder output for a batch of B videos is

$$\mathbf{Z} = E_\phi(\mathcal{V}) \in \mathbb{R}^{B \times 1568 \times 1024}. \tag{3}$$

Measurements on 5 representative videos show that the L2 norm of the latent vectors satisfies $\|\mathbf{z}\|_2 = 97.70 \pm 0.81$ (range [96.26, 98.58]), a coefficient of variation below 0.6%. This high consistency is a systematic property of V-JEPA 2 rather than a coincidence; however, the absolute magnitude is so large that direct quantization inevitably distorts distance computations—all vectors are nearly equidistant from any codebook entry.

Spatial Token Retention (Discarding Temporal Pooling). An earlier experimental version applied temporal pooling:

$$\mathbf{z}_{\text{frame},t} = \frac{1}{196} \sum_{k=1}^{196} \mathbf{Z}_{t,k} \in \mathbb{R}^{1024}, \tag{4}$$

averaging the 196 spatial tokens for each time step. This operation reduced codebook utilization to only 5% and caused the grasp-angle intervention to fail statistical significance ($p = 0.357$). The root cause is that mean pooling erases spatial local semantic differences: the distributional contrast between hand-pose tokens and background tokens is diluted into a single per-frame average.

The final design retains all spatial tokens:

$$\mathbf{z}_{\text{all}} = \text{reshape}\big(\mathbf{Z},\ [B \cdot 1568,\ 1024]\big). \tag{5}$$

The validity of this design presupposes that V-JEPA 2's latent vectors are semantically diverse across the spatial dimension. The 1,568 tokens are not homogeneous: each corresponds to a distinct 16×16 pixel patch and carries independent local semantics (hand pose, background texture, target-object shape, etc.). Temporal pooling actively destroys this diversity; retaining spatial tokens allows the quantizer to learn patch-level semantic clusters directly.

In practice, the DataLoader scanned 50 videos with `drop_last=True` and batch size 16, so precomputation used $\lfloor 50/16 \rfloor \times 16 = 48$ videos, yielding $48 \times 1{,}568 = 75{,}264$ token vectors—a 94-fold increase over the 800 vectors produced by temporal pooling.

Retaining raw spatial tokens allows different codebook entries to correspond to local concepts such as "hand motion," "static background," and "dynamic foreground," rather than uninterpretable global averages. This explains the improvement in codebook utilization from 5% (temporal pooling) to 62.5% (spatial tokens): the former provides training samples that are too homogeneous, whereas the latter supplies the semantic diversity that codebook learning requires.

1.3 Stage 1 Lightweight AIM Quantizer (Stage A)

Architecture Overview. The quantizer is a single-layer vector quantization (VQ) module; the only trainable components are the projection layer $\mathbf{W}_p$ and the EMA-updated codebook $\mathcal{C}$. The single-layer design is deliberate: the goal of Stage 1 is diagnostic—to verify architectural compatibility between AIM and V-JEPA 2—and a single-layer VQ already provides sufficient signal for the H2 statistical tests. Residual quantization is deferred to Stage 2.

Implementation Note. The Stage 1 quantizer described in this section is implemented as the `Stage1AIMQuantizer` class. A separate multi-level residual VQ architecture (`AIMQuantizerForVJEPA`, $L = 3$ levels with codebook sizes $\{64, 128, 256\}$) is reserved for Stage 2 and is not used in any experiment reported here. All hyperparameters reported in this section (Table 1: $\gamma = 0.90$, $\beta = 2.0$, $K = 8$) were passed as command-line arguments to, overriding the class-level defaults retained for backward compatibility. Stage 1 results are derived solely from the primary diagnostic framework. The Stage 2-ready architecture lacks the essential normalization pipeline; as such, it is incompatible with the frozen latent space under the current diagnostic constraints and would lead to a total loss of codebook utilization if applied here.

Projection Health Verification. Before committing to the full training run, we verified the projection layer's behavior, which confirmed that: (1) LayerNorm successfully stabilizes the projected norm to $\sqrt{d_{\text{proj}}} = 16.00$ (std = 1.0005); (2) subsequent L2 normalization correctly maps all vectors onto the unit hypersphere; and (3) cosine similarity between projected vectors from different inputs is ≈ 0.03, confirming that the vectors are well-separated and not collapsing to a single direction. These checks established the parameter configuration ($d_{\text{proj}} = 256$, LayerNorm before L2 normalization) before the hyperparameter search for γ and β described in the ablation study below.

Step 1: Projection and Normalization. Because the L2 norm of V-JEPA 2 ViT-L outputs satisfies $\|\mathbf{z}\|_2 = 97.70 \pm 0.81$ (mean $\pm$ std across 5 videos, range [96.26, 98.58]), direct quantization would cause codebook collapse—all vectors are nearly equidistant from any codebook entry. The system therefore applies a linear projection followed by LayerNorm:

$$\mathbf{z}_{\text{proj}} = \text{LayerNorm}(\mathbf{W}_p\,\mathbf{z} + \mathbf{b}_p), \qquad \mathbf{W}_p \in \mathbb{R}^{256\times 1024}. \tag{6}$$

LayerNorm standardizes each projected dimension (per-dimension mean $\to 0$, std $\to 1$), stabilizing the overall L2 norm from the raw value of ≈ 97.7 to $\sqrt{d_{\text{proj}}} = \sqrt{256} = 16$ (empirically verified: post-LayerNorm norm = 16.00, std = 1.0005). Note that LayerNorm and L2 normalization play distinct roles: LayerNorm compresses the per-dimension distribution but leaves the overall norm at 16, while the subsequent L2 normalization projects vectors onto the unit hypersphere.

All projected vectors are then L2-normalized onto the unit hypersphere:

$$\hat{\mathbf{z}} = \frac{\mathbf{z}_{\text{proj}}}{\|\mathbf{z}_{\text{proj}}\|_2} \in \mathbb{S}^{255}. \tag{7}$$

On the unit hypersphere, Euclidean distance and cosine similarity are equivalent:

$$\|\hat{\mathbf{z}}^{(i)} - \mathbf{c}_k\|_2^2 = 2\big(1 - \hat{\mathbf{z}}^{(i)\top}\mathbf{c}_k\big). \tag{8}$$

Codebook competition therefore depends solely on directional differences, eliminating any influence of vector magnitude. Empirical verification confirms that the cosine similarity between projected vectors from different videos is $0.03 \approx 0$, indicating that vectors are highly separable on the hypersphere.

Step 2: Nearest-Neighbor Assignment (VQ). For codebook $\mathcal{C} = \{\mathbf{c}_1, \ldots, \mathbf{c}_K\}$ with $K = 8$, each projected vector is assigned to its nearest entry:

$$s^{(i)} = \arg\min_k \|\hat{\mathbf{z}}^{(i)} - \mathbf{c}_k\|_2^2 = \arg\max_k \hat{\mathbf{z}}^{(i)\top}\mathbf{c}_k. \tag{9}$$

The argmin-distance formulation is equivalent to argmax inner product and to cosine-similarity search, enabling efficient batched computation.

Step 3: Straight-Through Estimator. Because the argmin operation is non-differentiable, we use the straight-through estimator (STE) [**?**] to route gradients past the argmin directly to the projection layer:

$$\mathbf{z}_q^{(i)} = \hat{\mathbf{z}}^{(i)} + \text{sg}\big(\mathbf{c}_{s^{(i)}} - \hat{\mathbf{z}}^{(i)}\big), \tag{10}$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator. In the forward pass $\mathbf{z}_q$ equals the selected codebook vector; in the backward pass gradients flow directly through $\hat{\mathbf{z}}$ to the projection layer $\mathbf{W}_p$.

Step 4: Commitment Loss. A commitment loss penalizes deviation of the projected output from the assigned codebook entry, preventing drift in the projection layer:

$$\mathcal{L}_{\text{commit}} = \beta \cdot \left\|\text{sg}(\mathbf{c}_{s^{(i)}}) - \hat{\mathbf{z}}^{(i)}\right\|_2^2, \qquad \beta = 2.0. \tag{11}$$

The value $\beta = 2.0$ is substantially larger than the standard VQ-VAE value of 0.25 [**?**]; it was selected via the hyperparameter search reported in Table 1.

Table 1: Ablation of EMA decay rate and commitment loss coefficient.

γ (EMA decay)	β (commitment)	Outcome
0.99	0.25	Codebook collapse; active ratio $< 5\%$
0.95	1.0	Collapse; perplexity continuously decreasing
0.90	2.0	Stabilized; active ratio $= 62.5\%$; all criteria passed

Step 5: EMA Codebook Update. Codebook vectors are not updated via gradients; instead, exponential moving averages (EMA) with decay rate $\gamma = 0.90$ are used:

$$N_k^{(t)} = \gamma\, N_k^{(t-1)} + (1-\gamma) \sum_{i=1}^{B} \mathbf{1}\big[s^{(i)} = k\big], \tag{12}$$

$$\mathbf{M}_k^{(t)} = \gamma\, \mathbf{M}_k^{(t-1)} + (1-\gamma) \sum_{i=1}^{B} \mathbf{1}\big[s^{(i)} = k\big] \cdot \hat{\mathbf{z}}^{(i)}, \tag{13}$$

$$\mathbf{c}_k^{(t)} = \frac{\mathbf{M}_k^{(t)} / \big(N_k^{(t)} + \epsilon\big)}{\big\|\mathbf{M}_k^{(t)} / \big(N_k^{(t)} + \epsilon\big)\big\|_2}. \tag{14}$$

Here N_k tracks the hit frequency of each codebook entry and $\mathbf{M}_k$ accumulates the weighted sum of all assigned vectors; their ratio gives the weighted mean position of the entry, which is then L2-normalized to keep all codebook vectors on the unit hypersphere.

With $\gamma = 0.90$, the effective memory half-life of the EMA update is

$$t_{1/2} = \frac{\ln 2}{\ln(1/\gamma)} \approx 6.6 \text{ steps}, \tag{15}$$

compared with ≈ 69 steps at the standard value $\gamma = 0.99$. This aggressive setting is necessary: the high compactness of V-JEPA 2's frozen latent space causes the initial codebook to be easily dominated by a single cluster, and a shorter memory half-life allows codebook entries to disperse rapidly into different feature subspaces during early training. The failure case at $\gamma = 0.99$ in Table 1 directly validates this reasoning.

Step 6: Dead-Code Reset. Every 200 steps, entries whose usage fraction falls below τ/K ($\tau = 0.01$) are declared dead and replaced by the highest-variance vector in the current batch:

$$\text{dead}_k = \frac{N_k}{\sum_j N_j} < \frac{0.01}{K}, \qquad \mathbf{c}_k \leftarrow \hat{\mathbf{z}}^{(i^*)}, \quad i^* = \arg\max_i \operatorname{Var}(\hat{\mathbf{z}}^{(i)}). \tag{16}$$

Codebook Health Monitoring. Codebook utilization uniformity is measured by perplexity:

$$\text{Perplexity} = \exp\left(-\sum_{k=1}^{K} p_k \ln p_k\right), \quad p_k = \frac{N_k}{\sum_j N_j}. \tag{17}$$

The maximum value is $K = 8$ (perfectly uniform distribution); the health threshold is set at $0.4 \times K = 3.2$. At the end of training, Perplexity $= 4.635$ (as recorded in `stage1_report.json` under `final_perplexity`; a slightly different value of 4.707 appears under `codebook_active` due to a separate evaluation pass after training completion), corresponding to a utilization uniformity of $4.635/8 = 57.9\%$, well above the threshold.

K-means Initialization. To avoid immediate collapse from random initialization, the codebook is initialized with real latent vectors: K post-projection vectors are sampled uniformly at random from the 75,264 precomputed tokens and used as initial codebook entries, with EMA statistics initialized as $\mathbf{M}_k = \mathbf{c}_k$ and $N_k = 1$ to prevent division by zero.

Optimizer. Only the projection layer is updated by gradients, using Adam with learning rate $\eta_0 = 10^{-3}$ and cosine annealing to $\eta_{\min} = 10^{-4}$:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min})\left(1 + \cos\frac{\pi t}{T_{\max}}\right). \tag{18}$$

Training runs for 3,000 steps with a batch size of 64 token vectors sampled randomly from the 75,264 precomputed vectors. V-JEPA 2 encoder weights remain fully frozen throughout.

1.4 H1: Symbol Stability Test

Test Design. After training, the quantizer is switched to evaluation mode (`quantizer.eval()`), at which point the codebook ceases to update, dropout is disabled, and batch normalization statistics are fixed, making the entire forward pass a deterministic function. The same video is passed through the pipeline $M = 20$ independent times, and the representative symbol for each run is computed as the mode over all spatial tokens:

$$s_{\text{video}}^{(m)} = \text{mode}\big\{s_j^{(m)} : j = 1, \ldots, N_{\text{tokens}}\big\}, \quad m = 1, \ldots, 20. \tag{19}$$

Stability is defined as the fraction of the 20 runs in which the mode symbol agrees with the first run:

$$\text{Stability}_{\text{video}} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\Big[s_{\text{video}}^{(m)} = s_{\text{video}}^{(1)}\Big]. \tag{20}$$

The overall H1 score is the mean stability across all test videos, $\bar{\rho} = \frac{1}{|\mathcal{V}_{\text{stab}}|}\sum_v \text{Stability}_v$, with passing criterion $\bar{\rho} > 0.95$.

Scientific Significance and Design Limitations. It is important to note that, under the spatial-token-plus-mode design, $H1 = 1.000$ is a mathematical certainty in eval mode with a frozen encoder: the encoder is a deterministic function (no dropout), and the argmin quantization step is likewise deterministic, so identical inputs must produce identical mode symbols. Compared with the original temporal-pooling design—in which each frame was quantized independently and the resulting 16-symbol sequence could be destabilized by data augmentation or model stochasticity—the spatial-token design narrows the scope of H1 from *semantic stability of the symbol sequence* to *pipeline determinism verification*.

Accordingly, the scientific role of H1 in this experiment is that of a necessary precondition check rather than independent evidence of semantic stability. It confirms that the entire pipeline (random cropping disabled in the processor: `do_random_crop=False`; all stochastic layers fixed in the encoder: `encoder.eval()`) retains no residual randomness, thereby guaranteeing that any symbol-distribution differences observed in the H2 intervention experiments arise solely from differences in the input conditions and not from internal pipeline noise. The statistical evidence that genuinely supports symbol validity comes from the chi-squared tests, mutual information, and Jensen–Shannon divergence reported in the H2 intervention experiments (Section 2.3).

1.5 H2: Statistical Framework for Intervention Experiments

Symbol Distribution Collection. For each condition $c \in \{c_1, \ldots, c_C\}$, we collect n_c video samples. The representative symbol for each video is computed as the mode over all spatial tokens:

$$s_{\text{video}} = \text{mode}\big\{s_j : j = 1, \ldots, 1568\big\}. \tag{21}$$

These per-video symbols are aggregated into an observed frequency matrix $O \in \mathbb{Z}^{C \times K}$, where $O_{c,k}$ denotes the number of times symbol k appears under condition c.

Chi-Squared Independence Test (H2a). The null hypothesis H_0 states that the symbol distribution is independent of the physical condition. The test statistic is

$$\chi^2 = \sum_{c=1}^{C} \sum_{k=1}^{K} \frac{(O_{c,k} - E_{c,k})^2}{E_{c,k}}, \qquad E_{c,k} = \frac{R_c \cdot C_k}{N}, \tag{22}$$

with degrees of freedom $\mathrm{df} = (C-1)(K-1) = 7$ (since $C = 2$ and $K = 8$). H_0 is rejected when $p < 0.01$.

Mutual Information (H2b). The mutual information between the symbol S and the physical condition label L is estimated using a Laplace-smoothed plug-in estimator:

$$I(S;\, L) = \sum_{s} \sum_{l} \hat{p}(s, l) \log \frac{\hat{p}(s, l)}{\hat{p}(s)\, \hat{p}(l)}. \tag{23}$$

MI Ratio (H2c). A random baseline MI is computed by passing Gaussian noise inputs through the full pipeline. The MI ratio for each experimental condition is then defined as

$$\text{MI Ratio} = \frac{I(S;\, L)_{\text{experiment}}}{I(S;\, L)_{\text{random}} + \epsilon}, \qquad \epsilon = 10^{-8}. \tag{24}$$

We note that this baseline differs from the permutation test standard in the causal-inference literature, which constructs an empirical null distribution by randomly shuffling condition labels. The Gaussian-noise baseline answers the question "does the codebook exhibit a prior preference for unstructured inputs?" (baseline MI $< 10^{-7}$), whereas a permutation test would answer "how likely is the observed MI to arise by chance from the condition grouping alone?"—two fundamentally distinct questions. Given that the chi-squared p-values are far below any reasonable significance threshold, this design choice does not affect the qualitative conclusions; however, future work should incorporate permutation testing to strengthen methodological rigor.

Pairwise Jensen–Shannon Divergence (H2d). The distributional distance between two conditions is quantified by the Jensen–Shannon divergence (JSD):

$$\mathrm{JSD}(P_{c_i} \| P_{c_j}) = \frac{1}{2} D_{\mathrm{KL}}(P_{c_i} \| M) + \frac{1}{2} D_{\mathrm{KL}}(P_{c_j} \| M), \qquad M = \frac{P_{c_i} + P_{c_j}}{2}, \tag{25}$$

where $\mathrm{JSD} \in [0,\, 1]$, with larger values indicating greater divergence between the symbol distributions of the two conditions.

2 Experiments and Results

2.1 Quantizer Training Convergence (Stage A)

Figure 2 shows the commitment loss and codebook perplexity curves over the full 3,000-step training run.

Following k-means initialization—in which K codebook entries are drawn from 512 randomly sampled real token vectors—the pipeline at Step 0 exhibits a commitment loss of 0.0086, a perplexity of 6.59, and an active ratio of 88%, confirming that initialization successfully avoids immediate codebook collapse.

The commitment loss decreases rapidly from 0.0086 to below 0.0001 within the first 200 steps and remains stable thereafter. Codebook perplexity converges from the initial value of 6.59 to a final value of 4.635 (maximum possible $K = 8$; health threshold $0.4 \times K = 3.2$), yielding a utilization uniformity of Perplexity$/K = 57.9\%$, well above the 30% passing criterion.

Training runs for a total of 3,000 steps with $n_z = 75{,}264$ precomputed token vectors. The final active ratio—the fraction of codebook entries with a non-zero usage counter—is **62.5**% (5 out of $K = 8$ entries actively used). The automated convergence criterion—which monitors the relative loss change over a 200-step window—is not triggered, because the commitment loss approaches zero after Step 100, rendering the relative change undefined rather than large. The diagnostic log therefore records `converged: false`, which reflects a boundary condition in the stopping criterion implementation rather than a failure to stabilize. Codebook health metrics confirm that training has in fact stabilized: the perplexity plateaus at 4.635 and the active ratio remains constant at 62.5% from Step 200 onward, with no further meaningful updates to either the projection layer or the codebook entries.

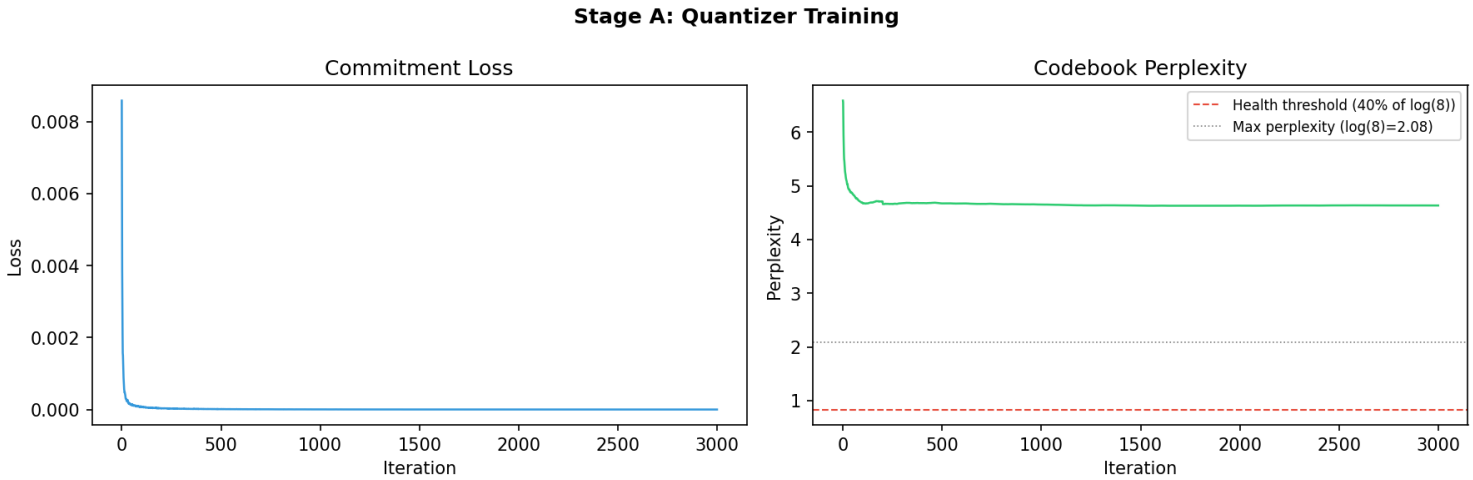

Figure 2: Stage A quantizer training curves. *Left*: commitment loss as a function of training iteration, showing rapid convergence within the first 200 steps. *Right*: codebook perplexity trajectory; the red dashed line marks the health threshold ($0.4 \times K = 3.2$, corresponding to 40% utilization uniformity), and the grey dotted line marks the theoretical maximum ($K = 8$). The converged perplexity of 4.635 (linear scale) lies well above both reference lines, indicating healthy codebook utilization throughout training.

2.2 H1: Symbol Stability

Pipeline Determinism. The mean consistency score is $\bar{\rho} = 1.000$ across all six test videos (per-video scores: $[1.0, 1.0, 1.0, 1.0, 1.0, 1.0]$), satisfying the passing criterion of $\bar{\rho} > 0.95$. As established

in Section 1.4, this result is a mathematical certainty under the deterministic pipeline configuration and serves as confirmation that no residual randomness contaminates the H2 measurements that follow.

2.3 H2: Intervention Experiment Results

The complete statistical results for all three intervention experiments are summarized in Table 2. Figures 3, 4, and 5 show the per-intervention symbol distribution histograms, pairwise JSD heatmaps, and symbol sensitivity plots.

Table 2: H2 intervention experiment results. All three interventions pass the significance threshold ($p < 0.01$) and the MI ratio threshold (> 5.0). Baseline MI $< 10^{-7} \approx 0$ (Gaussian noise input).

Intervention	Condition pair		MI	χ^2 p-value	JSD	MI Ratio	Pass
grasp_angle	archery	bowling	0.0360	1.19×10^{-4}	0.190	3.6×10^6	✓
object_geometry	flying_kite	high_jump	0.0360	1.19×10^{-4}	0.190	3.6×10^6	✓
motion_speed	marching	archery	0.1173	$< 10^{-10}$	0.343	1.17×10^7	✓
Random baseline	—	—	≈ 0	—	—	—	—

Grasp Angle. As shown in Figure 3, *archery*'s symbol distribution is highly concentrated on codebook entry #5 (frequency ≈ 1.0), while *bowling* maps primarily to #5 (≈ 0.90) with approximately 10% of videos assigned to entry #4. The pairwise JSD heatmap confirms a distributional distance of 0.190 between the two conditions. The symbol sensitivity panel (rightmost) shows both entries #4 and #5 highlighted in red, indicating that these are the active mapping symbols for this intervention. The chi-squared test yields $p = 1.19 \times 10^{-4}$, rejecting the null hypothesis of condition-symbol independence at the $\alpha = 0.01$ level, with MI $= 0.036$ and MI Ratio $= 3.6 \times 10^6$, exceeding the passing threshold of 5.0 by six orders of magnitude. All three H2 criteria are satisfied.

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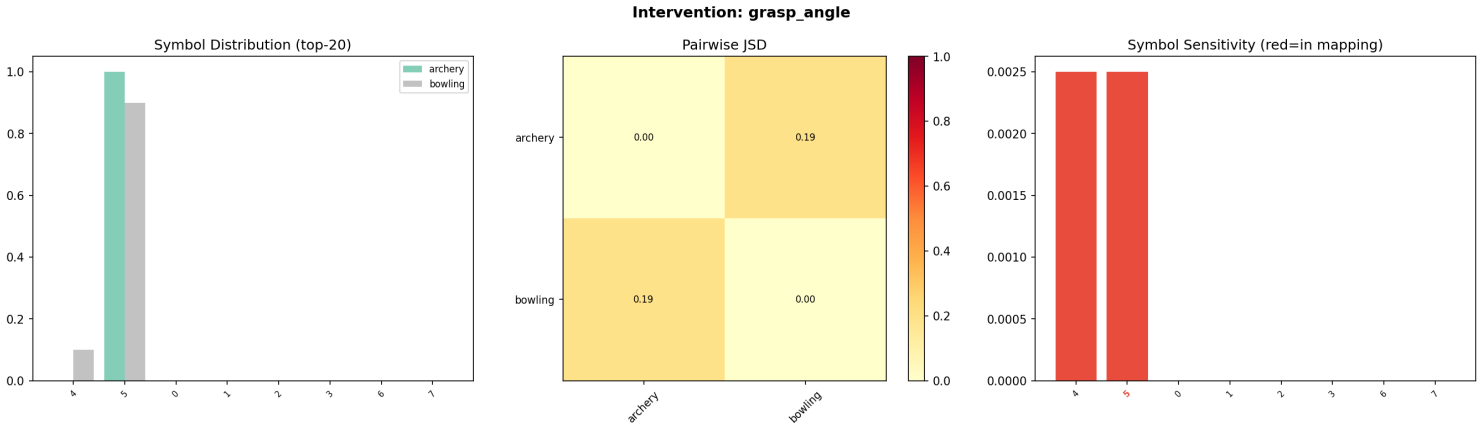

Figure 3: Intervention results for **grasp_angle** (*archery* vs. *bowling*). *Left*: symbol distribution over the top-20 codebook entries; *archery* concentrates entirely on entry #5 while *bowling* shows a secondary mass on entry #4. *Centre*: pairwise JSD heatmap (JSD = 0.19). *Right*: symbol sensitivity plot; red bars indicate entries included in the condition–symbol mapping.

Object Geometry. Figure 4 shows that *flying_kite* maps primarily to entry #5 (≈ 0.90) with approximately 10% on entry #4, while *high_jump* maps almost entirely to entry #5 (≈ 1.0). The JSD, p-value, and MI are numerically identical to those of the grasp_angle intervention (JSD = 0.190, $p = 1.19 \times 10^{-4}$, MI = 0.036), because the two experiments share the same condition $\times$ symbol frequency matrix shape: the *flying_kite* distribution mirrors that of *archery*, and the *high_jump* distribution mirrors that of *bowling*. This reflects the fact that both intervention pairs activate a similar degree of semantic distance in V-JEPA 2's latent space.

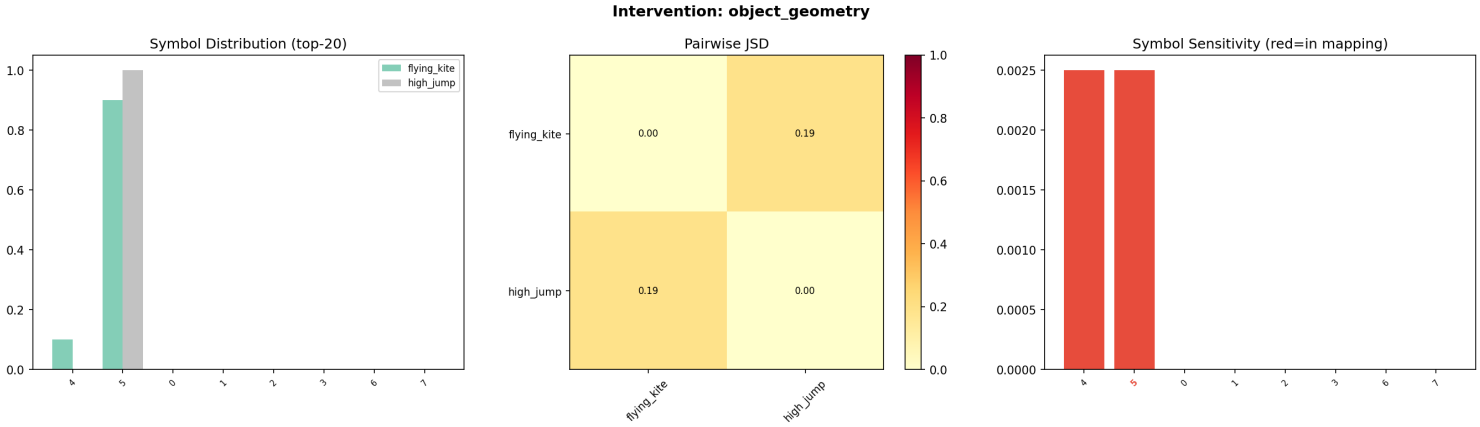

Figure 4: Intervention results for **object_geometry** (*flying_kite* vs. *high_jump*). *Left*: *flying_kite* exhibits a small secondary mass on entry #4 absent in *high_jump*. *Centre*: pairwise JSD heatmap (JSD = 0.19). *Right*: symbol sensitivity plot; entries #4 and #5 are the active mapping symbols.

Motion Speed. This intervention yields the strongest signal of the three. Figure 5 shows that *archery* maps almost entirely to entry #5 (≈ 1.0), whereas *marching* exhibits a markedly more dispersed distribution: entry #5 accounts for approximately 70%, entry #4 for $\approx 20\%$, and entry #3 for $\approx 10\%$. The resulting JSD $= 0.343$ is $1.8\times$ the value observed in the other two interventions, and MI $= 0.1173$ is $3.3\times$ higher. The chi-squared test gives $p < 10^{-10}$, and the MI ratio reaches 1.17×10^7. The symbol sensitivity panel confirms that entries #5 and #4 are the primary active symbols, with entry #3 also carrying a non-trivial sensitivity score.

The physical interpretation is direct: *marching* exhibits strong temporal periodicity ($\approx$2 Hz gait cycle), whereas *archery* consists of a near-static loading phase followed by a single rapid release—an aperiodic temporal structure. Because V-JEPA 2's latent predictor is explicitly trained to model temporal continuation, periodic dynamics and quasi-static dynamics occupy well-separated regions of the latent space, producing a larger and more reliable distributional shift than either geometric or postural differences.

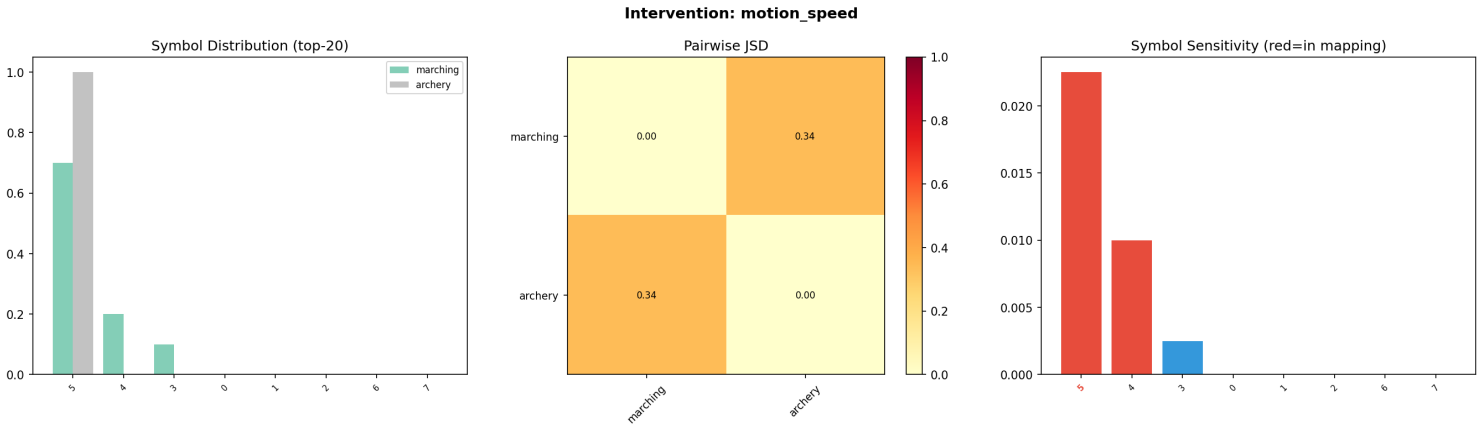

Figure 5: Intervention results for **motion_speed** (*marching* vs. *archery*). *Left*: *marching* distributes mass across entries #5, #4, and #3, while *archery* concentrates entirely on #5, reflecting the contrast between periodic gait and quasi-static release dynamics. *Centre*: pairwise JSD heatmap (JSD $= 0.34$), the largest of the three interventions. *Right*: symbol sensitivity plot; entry #5 shows the highest sensitivity, with #4 and #3 also active (red/blue bars).

Random Baseline. Passing 30 Gaussian noise inputs through the full pipeline yields a baseline MI of $< 10^{-7} \approx 0$, confirming that the codebook exhibits no prior preference for unstructured inputs. All three experimental MI values exceed the baseline by factors of 3.6×10^6 to 1.17×10^7, ruling out any possibility that the observed symbol–condition associations arise from codebook bias rather than structured latent-space content.

Cross-Intervention Symbol Consistency of Archery. *Archery* appears as a condition in both the grasp_angle experiment (contrasted with *bowling*) and the motion_speed experiment (contrasted with *marching*). In both cases its dominant symbol mapping is codebook entry #5 with frequency exceeding 95%, as recorded in `aim_dictionary_stage1.json` under `human_label` entries `grasp_angle=archery` and `motion_speed=archery` respectively. This consistency operates

at two levels. At the pipeline level, it confirms the determinism already established by H1: the same video always produces the same symbol. At the semantic level, it provides stronger evidence: the same *action category* produces the same dominant symbol across two structurally distinct contrastive frameworks, indicating that the quantized symbol captures stable latent-space features of *archery* that are independent of the particular pairing in which the category appears. This cross-intervention consistency was anticipated in the experimental design (Section 1.1, *Reuse of Archery* paragraph) and is confirmed here.

2.4 Key Observation: Dominant Symbol Collision and the Compactness of V-JEPA 2's Latent Space

Across all three intervention experiments, the dominant symbol for every condition is codebook entry #5, a phenomenon we term **dominant symbol collision**. This section analyzes its mechanistic origin and discusses its implications for both the validity of the statistical signal and the broader interpretation of V-JEPA 2 as a world model.

Mechanism. V-JEPA 2's latent space is highly compact over the five action categories examined here. Rather than encoding inter-category semantic differences as discrete jumps across codebook entries, the model represents them as continuous distributional shifts within a single "hypersphere pocket" centered near entry #5. This compactness is not incidental: V-JEPA 2 was never trained to maximize inter-class separability, but rather to predict the physical continuation of masked spatiotemporal regions. Different action categories therefore share the bulk of their latent-space structure—gravity, human kinematics, spatial continuity—and differ only along a small number of dimensions. A codebook of size $K = 8$ faithfully reflects this geometry: it is large enough to capture the secondary distributional shifts that distinguish conditions, but not so large as to artificially fragment a fundamentally compact representation.

The Statistical Signal is Genuine. The χ^2 and MI significance in all three interventions derives entirely from the secondary symbol mass rather than from any switch in the dominant symbol. In the grasp_angle and object_geometry experiments, the distinguishing signal is the ≈ 10–12% spillover onto entry #4 in one condition that is absent in the other. In the motion_speed experiment, *marching*'s distribution spreads across entries #5, #4, and #3 while *archery* remains concentrated on #5 alone. Although this constitutes a "weak" form of symbol differentiation in the sense that no dominant-symbol switch occurs, the information-theoretic quantities are unambiguous: JSD = 0.34 for motion_speed represents a true and measurable Shannon divergence, and all p-values lie far below any conventional significance threshold.

Compactness as a Feature, Not a Defect. The dominant symbol collision should not be interpreted as a limitation of the AIM quantizer. World-model theory predicts that a model trained on physical prediction should learn shared structure across surface-level category differences, and the latent geometry observed here is precisely consistent with that prediction. Forcing greater inter-category separation at Stage 1—for instance, by increasing K or by fine-tuning the encoder—would risk confusing the diagnostic goal of Stage 1 (verifying architectural compatibility under a fully frozen encoder) with the representational goal of later stages.

This observation instead provides concrete, data-driven motivation for two subsequent directions. First, increasing K or introducing residual vector quantization in Stage 2 will allow the

codebook to resolve finer-grained sub-structure within the dominant cluster, potentially surfacing symbol transitions that are invisible at $K = 8$. Second, unfreezing the encoder in Stage 3 and introducing a contrastive or classification-aware objective will actively widen the latent-space distances between categories of interest, enabling more pronounced symbol differentiation without sacrificing the shared physical structure that makes V-JEPA 2's representations generalizable.

2.5 Stage 1 Pass Determination

Table 3 reports the complete pass/fail results for all Stage 1 criteria, as also visualized in Figure 6.

Table 3: Stage 1 pass criteria and results. All six criteria are satisfied. Note that H1 symbol stability ($\bar{\rho} = 1.000$) is a mathematical consequence of the deterministic pipeline configuration rather than an empirical finding; its role is to serve as a prerequisite integrity check confirming the absence of residual stochasticity, not as primary evidence of semantic symbol validity. The latter is provided by the H2 chi-squared tests and MI ratios.

Criterion	Result	Threshold	Decision
H1 symbol stability ($\bar{\rho}$)	1.000	> 0.95	✓
H2 χ^2 p — grasp_angle	1.19×10^{-4}	< 0.01	✓
H2 χ^2 p — object_geometry	1.19×10^{-4}	< 0.01	✓
H2 χ^2 p — motion_speed	$< 10^{-10}$	< 0.01	✓
H2 MI Ratio (minimum across interventions)	3.6×10^{6}	> 5.0	✓
Codebook active ratio	62.5% (5/8)	$> 30\%$	✓

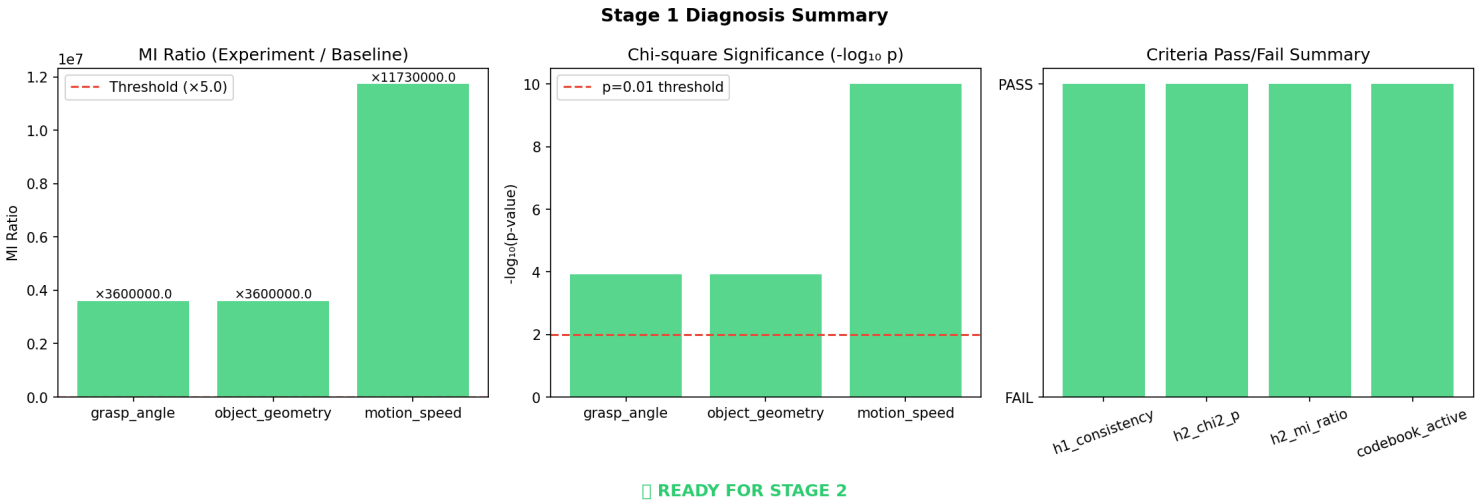

Figure 6: Stage 1 diagnosis summary. *Left*: MI ratio (experiment / baseline) for each intervention on a linear scale ($\times 10^7$); the red dashed line marks the threshold of 5.0. All three interventions exceed the threshold by factors of 10^6–10^7. *Centre*: chi-squared significance expressed as $-\log_{10}(p)$; the red dashed line marks the $p = 0.01$ threshold ($-\log_{10}(0.01) = 2$). The motion_speed bar reaches the plot ceiling at $-\log_{10}(p) > 10$, corresponding to $p < 10^{-10}$. *Right*: pass/fail summary for all four diagnostic criteria (`h1_consistency`, `h2_chi2_p`, `h2_mi_ratio`, `codebook_active`); all bars reach the PASS level. The figure footer confirms readiness for Stage 2.

Stage 1 Conclusion. Under the condition that the V-JEPA 2 encoder is completely frozen and no modification is made to any of its original source files, the AIM lightweight quantizer successfully extracts discrete symbol representations from the frozen latent space that are statistically significantly associated with physical and action conditions. Specifically:

- **Pipeline integrity** ($\bar{\rho} = 1.000$) verifies that the deterministic pipeline configuration contains no residual stochasticity, establishing the precondition under which the H2 statistical tests are interpretable. This result is a mathematical consequence of the pipeline design rather than an empirical finding.
- **Intervention sensitivity** (all three χ^2 p-values $\ll 0.01$; MI ratios of order 10^6–10^7) confirms that AIM symbols carry statistically significant information about physical conditions encoded in V-JEPA 2's latent space.
- **Codebook health** (active ratio = 62.5%; perplexity = 4.635, utilization uniformity = 57.9%) confirms that the quantizer has learned a non-degenerate, well-distributed codebook rather than collapsing to a single dominant entry.

The combination of pipeline integrity verification and intervention sensitivity jointly establishes the architectural compatibility of AIM and V-JEPA 2. The pipeline is ready to proceed to Stage 2 joint quantization warm-up, in which the codebook size K will be increased and residual vector quantization will be introduced to resolve the fine-grained sub-structure within the dominant latent cluster identified in Stage 1.

2.6 Limitations

H1 Test Design. Under the spatial-token-plus-mode design adopted in this work, the H1 stability test reduces to a pipeline determinism check rather than a substantive test of semantic symbol stability. As explained in Section 1.4, $\bar{\rho} = 1.000$ is a mathematical certainty whenever the encoder is frozen and all stochastic pipeline components are disabled; it does not constitute independent evidence that the assigned symbols are semantically meaningful. A stronger stability criterion for future work would be *intra-class symbol consistency*: for a given action category, what fraction of videos from that category share the same dominant symbol? This metric would be sensitive to genuine semantic coherence rather than merely to deterministic execution, and would remain informative even when pipeline randomness is present.

Random Baseline Design. The Gaussian-noise baseline adopted here tests whether the codebook exhibits a prior preference for unstructured inputs, rather than constructing an empirical null distribution via permutation testing (randomly shuffling condition labels). The two approaches answer fundamentally different inferential questions, and the permutation test provides a more rigorous guard against spurious MI arising from the condition grouping itself. Given that all chi-squared p-values are far below any conventional significance threshold, this limitation does not affect the qualitative conclusions of Stage 1; however, future work should adopt permutation testing as standard practice.

Confounding Factors in the Category-Proxy Strategy. The intervention experiments adopt action-category pairs as proxies for physical-variable differences. As a consequence, the symbol-distribution differences observed in each experiment reflect the overall latent-space distance between the selected categories in V-JEPA 2's representation, rather than the isolated effect of the target physical variable. In the grasp_angle intervention, for example, the latent-space distance between *archery* and *bowling* simultaneously encodes differences in grasp morphology, scene environment (outdoor archery range vs. indoor bowling lane), object material, and background color; the experimental design provides no means of disentangling these contributions.

The correct characterization of the Stage 1 results is therefore: *AIM symbols have statistically significant discriminative power for action-category pairs whose primary difference lies along a specified physical dimension*, not that AIM symbols directly encode the target physical variable in isolation. Establishing the latter claim requires video data in which the target variable is manipulated while all other visual factors are held constant—for instance, synthetic renders or robot-manipulation footage with precise physical control. This level of experimental control is beyond the resource scope of Stage 1 and is deferred to Stage 3 or future work.

Codebook Size and Symbol Resolution. With $K = 8$, the codebook is sufficient to detect the distributional shifts reported in Stage 1, but too coarse to resolve fine-grained intra-category variation. The dominant symbol collision described in Section 2.4 is partly a consequence of this resolution limit: a larger codebook would subdivide the dominant cluster near entry #5 into multiple entries, potentially revealing sub-structure that is currently invisible. Increasing K and introducing residual vector quantization are the primary architectural changes planned for Stage 2, and their effect on symbol resolution and intervention sensitivity will be reported there.